

11.1/11.2 Simple Linear Regression

The energy consumption per month of a chemical plant depends, among other things, on the average outside temperature.

So if X = the average outside temperature and Y = the energy consumption per month, then Y is **dependent** on X :

$$Y = f(X)$$

The variable X is called the **independent** variable or the **predictor** variable, Y is called the **dependent** variable or the **response** variable.

Determining the function f ("Is the relationship a straight line"?, "Does the relationship have a parabolic character?") is called **regression**.

A distinction is made between:

1. Simple and multiple regression

$$\left. \begin{array}{ll} X_1 & = \text{average outside temperature} \\ X_2 & = \text{number of production days per month} \\ Y & = \text{energy consumption per month} \end{array} \right\} \Rightarrow Y = f(X_1, X_2) \quad \text{twofold regression}$$

2. Linear and non-linear regression

$$\begin{array}{ll} Y = 3X_1 - 2X_2 + 4 & \text{(twofold) linear regression} \\ Y = -(X - 21)^2 + 2500 & \text{(simple) non-linear regression} \end{array}$$

Remark

In the case of **linear regression**, the **correlation coefficient** is used as a quantitative measure in answering the question of how well a scatter diagram is described by $Y = f(X)$. See Section 11.8.

Simple Linear Regression

Suppose:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

So the random variable Y depends linearly on x . The random variable ε ("epsilon") is called the **random error term**. This random error term includes reading errors, inaccuracy of the measuring instruments, etc.

We suppose

$$E(\varepsilon) = 0;$$

$$V(\varepsilon) = \sigma^2.$$

So

$$E(Y | x) = \beta_0 + \beta_1 x.$$

Using n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the **intercept** β_0 and the **slope** β_1 of the straight line will be estimated.

The least square estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ for β_0 and β_1

Given is a scatter plot with n observations (x_i, y_i) :

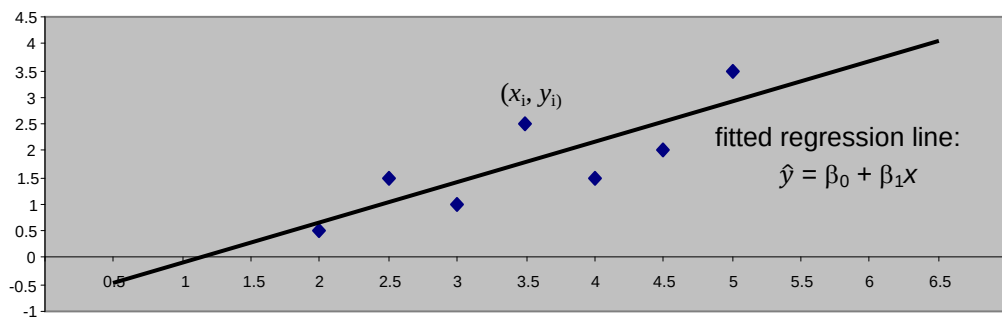


Figure 11.1 Scatter plot of y versus x

The following applies for the equation of the **fitted regression line** $\hat{y} = \beta_0 + \beta_1 x$, i.e. the straight line with **"intercept"** β_0 and **"slope"** β_1 :

"the sum of the squares of the deviations of the observations from the true regression line should be minimal".

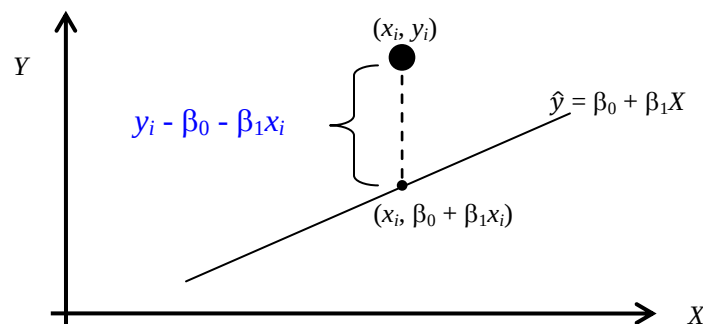


Figure 11.2 Deviations of the data from the estimated regression model

In other words, the function

$$L(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

should be minimal.

A necessary condition for this is:

$$\left. \begin{aligned} \frac{\partial L}{\partial \beta_0} \Big|_{\hat{\beta}_0, \hat{\beta}_1} &= 0 \\ \frac{\partial L}{\partial \beta_1} \Big|_{\hat{\beta}_0, \hat{\beta}_1} &= 0 \end{aligned} \right\} \Leftrightarrow \begin{aligned} \sum_{i=1}^n 2 \cdot (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^1 \cdot -1 &= 0 \\ \sum_{i=1}^n 2 \cdot (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^1 \cdot -x_i &= 0 \end{aligned}$$

After reduction, the so-called **least squares normal equations** arise:

$$\begin{aligned} \sum y_i &= n \cdot \hat{\beta}_0 + \hat{\beta}_1 \cdot \sum x_i & (1) \\ \sum x_i \cdot y_i &= \hat{\beta}_0 \cdot \sum x_i + \hat{\beta}_1 \sum x_i^2 & (2) \end{aligned}$$

Equation (2) - $\frac{\sum x_i}{n} \cdot$ Equation (1) gives:

$$\begin{aligned} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \cdot \bar{x} \\ \hat{\beta}_1 &= \frac{\sum x_i \cdot y_i - n \cdot \bar{x} \cdot \bar{y}}{\sum x_i^2 - n \cdot \bar{x}^2} \end{aligned}$$

The solution ($\hat{\beta}_0, \hat{\beta}_1$) is a **critical point** of $L(\beta_0, \beta_1)$. In the case of a least squares problem, a **minimum** occurs at ($\hat{\beta}_0, \hat{\beta}_1$).

Example 1

In a chemical plant, the average **outside temperature** (X, in °F) and **energy consumption** (Y) are recorded. Consider the data in Table 11.1 and the corresponding scatter plot in Figure 11.3.

Table 11.1 Outside temperature (X) and energy consumption (Y)

Month	X	Y	X·Y	X ²	Y ²
January	25	240	6000	625	57600
February	31	236	.	.	.
March	45	270	.	.	.
April	60	274	.	.	.
May	65	301	.	.	.
June	72	316	.	.	.
July	80	300	.	.	.
August	84	296	.	.	.
September	75	267	.	.	.
October	60	276	.	.	.
November	50	288	.	.	.
December	38	261	9918	1444	68121
Total	685	3325	193990	43245	927875

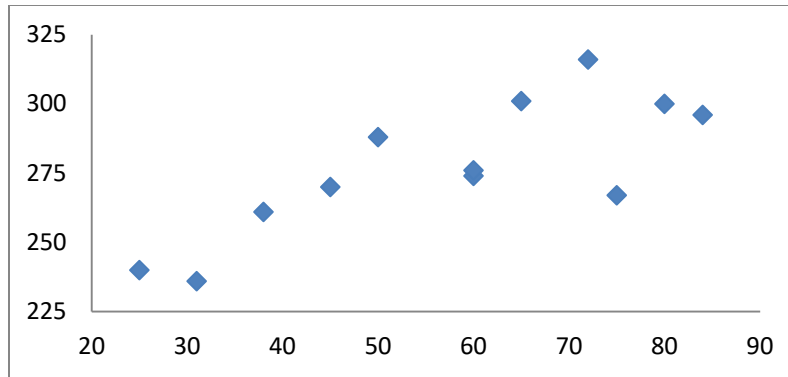


Figure 11.3 Outside temperature (X) and energy consumption (Y)

Entering the totals determined in Table 11.1 in the **least squares normal solutions** gives:

$$\hat{\beta}_1 = \frac{193990 - 12 \cdot \frac{685}{12} \cdot \frac{3325}{12}}{43245 - 12 \cdot \left(\frac{685}{12}\right)^2} = 1.010861913$$

$$\hat{\beta}_0 = \frac{3325}{12} - \hat{\beta}_1 \cdot \frac{685}{12} = 219.3799658$$

So

$$\hat{y} = 219.3800 + 1.0109x$$

Remarks

- In a month with an average outside temperature of 80 °F, the **expected / predicted** energy consumption is $219.3800 + 1.0108 \cdot 80 = 300.2489$, i.e.

$$\hat{y} = 300.2489$$

- The difference:

$$e_i = y_i - \hat{y}_i$$

is called the **residual**. The residual belonging to the month of July is therefore -0.2489 .

- It is also possible to determine the least squares estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ using a **graphical calculator**:

STAT→TESTS→LinRegTTest.

Choose at “RegEq” the function Y_1 .

vars/Y-vars/1/1

Check:

$$\begin{array}{llll} \hat{\beta}_0 & = & a & = & 219.3800 \\ \hat{\beta}_1 & = & b & = & 1.0109 \\ \hat{y}(80) & = & Y_1(80) & = & 300.2489188. \end{array}$$

The variance of the random error term

Predicting Y based on a given x_0 is more accurate if $\sigma^2 = V(\epsilon)$ is closer to zero. Information regarding the size of σ^2 is obtained by estimating σ^2 .

An **unbiased estimator for σ^2** (i.e. the variance of the **random error term**) is given by:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n e_i^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

This expression can be reduced to:

$$\hat{\sigma}^2 = \frac{\sum y_i^2 - \hat{\beta}_0 \cdot \sum y_i - \hat{\beta}_1 \cdot \sum x_i \cdot y_i}{n-2}$$

Example 2

See the data from Example 1: the relationship between *outside temperature* and *energy consumption*.

$$\hat{\sigma}^2 = \frac{927875 - 219.3740 \cdot 3325 - 1.0109 \cdot 193990}{12-2} = 233.9511$$

So

$$\hat{\sigma} = 15.2955$$

Remark

It is also possible to determine the size of $\hat{\sigma}$ using the option **LinRegTTest** of the **graphical calculator**.